

CSCI 432/532, Spring 2026

Homework 11

Due Monday, April 27 at 9:00am

Submission Requirements

- Type or clearly hand-write your solutions into a PDF format so that they are legible and professional. Submit your PDF on Canvas or Gradescope.
- Put each **sub**problem on a separate page and select the corresponding problem via the Gradescope interface.
- Do not submit your first draft. Type or clearly re-write your solutions for your final submission. If your submission is not legible, we will ask you to resubmit.

Collaboration and Resources

The *best* resources for completing the homework are (in no particular order):

- Your own notes from lectures and problem sessions, and/or lecture recordings.
- The lecture notes linked from the course website for the lecture day.
- The questions channel on Discord.
- Office hours.
- Discussions with classmates.

You may also access almost any resource in preparing your solution to the homework. The exception is that you may not seek answers to the problems on the homework directly, such as by pasting them into a GenAI tool, searching for solutions on a search engine, looking for them on Chegg or similar websites, or asking another person for the solution.

Additionally, you **must**

- Write your solutions in your own words—this means that you should never be *copying* anything of substance from any source, and
- credit every resource you use, including GenAI tools, by providing links or full chat transcripts. If you use the provided LaTeX template, you can use the `Sources` environment for this. Ask if you need help!

Remember, if you choose to use GenAI tools, not only must you provide a full transcript of your conversation (or a link to one), you must also use the following prompt (or some variation of it), and provide a full transcript of the conversation.

Prompt:

You are acting as a teaching assistant for an advanced undergraduate/graduate algorithms and computer science theory course.

Your role is not to provide full solutions or final answers.

Instead, you should act like we are having a conversation. Ask me a single question and let me respond. Avoid asking me many questions at once or giving me a lot of information at one time. Guide me using hints, leading questions, partial insights, and sanity checks. Help me debug my own proofs, algorithms, or reductions. Do not give a complete solution or full proofs. Be patient with me. Don't give me details so that I can move on to the next part of a problem. Make sure that I understood before moving on.

The course uses Jeff Erickson's Models of Computation lecture notes, so you should guide me to answer problems using the definitions, notation, and style from those notes.

Assume I am a serious student trying to learn, not to shortcut the assignment. I am attaching the assignment, so you should refuse to provide a complete solution to the problems listed. Of course, you can walk me through the "Solved Problems" listed at the end if I ask.

1. Do you think that all classical **reversible** operations also valid quantum operations? If not, give a countereample. If yes, explain why. Any reasonable attempt will get credit, so try to think about it yourself rather than looking up an answer!
2. Prove that the following quantum algorithm on two qubits is unitary by showing that it preserves the unitary property on $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$.

DOUBLEHADAMARD:

initialize two qubits A, B
put A and B into any basic state (all amplitude on one of the four states)
Hadamard A
Hadamard B

3. Explain why, given n qubits, the operation that applies Hadamard to each qubit puts equal amplitude on each of the 2^n basic states. You may provide either a formal proof or a less formal explanation.
4. Optional: watch the 3Blue1Brown video on quantum computing and Grover's algorithm:
<https://www.youtube.com/watch?v=RQWpF2Gb-gU>.